2.2 The Wave Equation

Let us first consider Maxwell's equations in vacuum, where the charge density is $\rho = 0$. Substituting the expression for the electric field $\boldsymbol{E}$ in (2.1.11) into the first equation of (2.1.37), we obtain

$$div\ \boldsymbol{E} = -div\ (grad\ \varphi) - \frac{\partial}{c\partial t}(\boldsymbol{\nabla} \cdot \boldsymbol{A}) = 0 \tag{2.2.1}$$

Alternatively, taking the curl of both sides of the first equation in (2.1.13) and substituting the second equation in (2.1.37), we obtain

$$\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \cdot \boldsymbol{E}) - \boldsymbol{\nabla^2 E} = -\frac{\partial^2 \boldsymbol{E}}{c^2 \partial t^2} \tag{2.2.2}$$

Substituting the expression for the magnetic field $\boldsymbol{H}$ in (2.1.11) into the fourth equation of (2.1.37), and then taking the curl of both sides of the resulting equation, we obtain

$$\boldsymbol{rot}\ (\boldsymbol{rot}\ \boldsymbol{H}) = -\frac{\partial}{c\partial t}\boldsymbol{rot}\left(grad\ \varphi + \frac{\partial \boldsymbol{A}}{c\partial t}\right) = -\frac{\partial^2 \boldsymbol{H}}{c^2 \partial t^2} \tag{2.2.3}$$

Expanding the double curl on the left-hand side of (2.2.3) and simplifying, we obtain

$$\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \cdot \boldsymbol{H}) - \boldsymbol{\nabla^2 H} = -\frac{\partial^2 \boldsymbol{H}}{c^2 \partial t^2} \tag{2.2.4}$$

By directly substituting the expressions for the electric field $\boldsymbol{E}$ and magnetic field $\boldsymbol{H}$ into the fourth equation of (2.1.37) and expanding the resulting expression, we obtain

$$\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \cdot \mathbf{A}) - \boldsymbol{\nabla^2 A} + \frac{\partial grad\ \varphi}{c\partial t} = -\frac{\partial^2 \boldsymbol{A}}{c^2 \partial t^2} \tag{2.2.5}$$

Further analysis requires us to consider the choice of gauge for the electromagnetic potentials. Two commonly used gauges in electromagnetism are the Coulomb gauge, defined by

$$\boldsymbol{\nabla} \cdot \boldsymbol{A} = \mathbf{0} \tag{2.2.6}$$

and the more general Lorenz gauge, defined by

$$\frac{\partial A^i}{\partial x^i} = 0 \tag{2.2.7}$$

Comparing (2.2.6) and (2.2.7), we can see that the Coulomb gauge is a special case of the Lorenz gauge, while the Lorenz gauge provides a more general formulation. In the

following discussion of electromagnetic fields, we will examine these gauges separately according to the physical context. Both of these gauges lead to the following result:

$$\boldsymbol{\nabla}\cdot\mathbf{A}+\frac{\partial\varphi}{c\partial t}=\boldsymbol{\nabla}\cdot\boldsymbol{E}=\boldsymbol{\nabla}\cdot\boldsymbol{H}=0 \tag{2.2.8}$$

Equation (2.2.8) is commonly referred to as the transversality condition for electromagnetic waves. Under this condition, every component of the equations (2.2.1)–(2.2.5) satisfies the same equation:

$$\boldsymbol{\nabla^2}f(\boldsymbol{r},t)=\frac{\partial^2 f(\boldsymbol{r},t)}{c^2\partial t^2} \tag{2.2.9}$$

Equation (2.2.9) is called the wave equation, also known as the d'Alembert equation. It can alternatively be written as

$$\square^2 f(\boldsymbol{r},t)=\frac{\partial^2 f(\boldsymbol{r},t)}{c^2\partial t^2}-\boldsymbol{\nabla^2}f(\boldsymbol{r},t)=0 \tag{2.2.10}$$

Here, $\square^2$ denotes the d'Alembert operator.

As an introductory example, let us consider waves propagating along a string. Assume that the tension is uniform throughout the string, and that the displacement of each point on the string is described by a function $y=f(x,t)$.

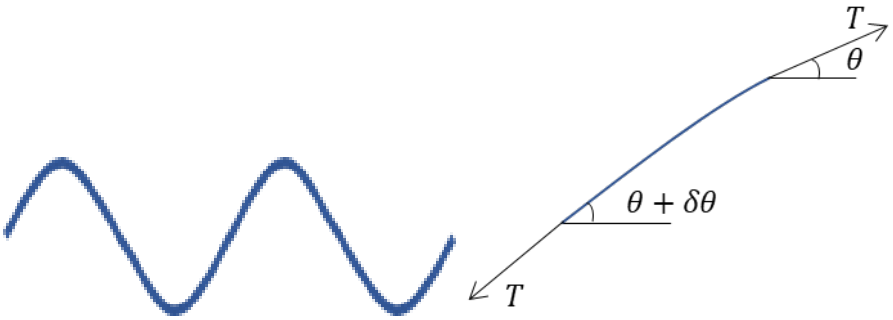


Fig.2.1 Illustration of a wave on a string and its microscopic structure

Analyzing the forces acting on this small segment, we obtain the following equation in the y-direction:

$$F=-T\cdot\sin(\theta+\delta\theta)+T\cdot\sin(\theta) \tag{2.2.11}$$

Suppose that the mass of this segment is m. The so-called wave motion can be understood as the simple harmonic motion of each material point in the y-direction. Treating this small segment as a point particle, Newton's second law gives

$$F=ma=m\frac{\partial^2 y}{\partial t^2}=m\frac{\partial^2 f}{\partial t^2} \tag{2.2.12}$$

To proceed, we need to introduce an additional assumption: the overall amplitude of the wave is sufficiently small, such that $\theta \approx 0$. We now examine the origin of the force F. We have

$$\begin{aligned} F &= -T \cdot \sin(\theta + \delta\theta) + T \cdot \sin(\theta) \\ &\approx -T \cdot \tan(\theta + \delta\theta) + T \cdot \tan(\theta) \\ &\equiv -T \cdot \left.\frac{\partial f}{\partial x}\right|_{x} + T \cdot \left.\frac{\partial f}{\partial x}\right|_{x+\delta x} = T \cdot \frac{\partial^2 f}{\partial x^2} \end{aligned} \tag{2.2.13}$$

Combining (2.2.12) and (2.2.13), we obtain

$$T \cdot \frac{\partial^2 f}{\partial x^2} = m \frac{\partial^2 f}{\partial t^2} \tag{2.2.14}$$

The above equation is the one-dimensional form of the wave equation. By introducing the Laplacian operator, we can generalize it to three dimensions:

$$\nabla^2 f = \mu \frac{\partial^2 f}{\partial t^2} \tag{2.2.15}$$

Here, μ is a constant determined by the properties of the medium and the initial conditions.

Finding solutions to the wave equation has been a subject of extensive study throughout the history of physics. For now, let us focus on the solution to the one-dimensional wave equation. The original equation is

$$K \frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 f}{\partial t^2} \tag{2.2.16}$$

Here, we adopt an approach based on the transport equation, which is relatively straightforward to understand and also provides a more intuitive physical interpretation. First, consider the following equation for a function $\varphi(x, y)$:

$$a \frac{\partial \varphi}{\partial x} + b \frac{\partial \varphi}{\partial y} = 0 \tag{2.2.17}$$

It is straightforward to verify that

$$\begin{cases} \dfrac{\partial \varphi}{\partial x} = b \\ \dfrac{\partial \varphi}{\partial y} = -a \end{cases} \tag{2.2.18}$$

satisfies the transport equation. Since $bx - ay = c$, the solution can be written as

$$\varphi(x, y) = \lambda(bx - ay) \tag{2.2.19}$$

where λ is an arbitrary function. Applying this approach to the wave equation, we

obtain

$$K\frac{\partial^2 f}{\partial x^2}=\frac{\partial^2 f}{\partial t^2}\Longrightarrow\left(\sqrt{K}\frac{\partial}{\partial x}-\frac{\partial}{\partial t}\right)\left(\sqrt{K}\frac{\partial}{\partial x}+\frac{\partial}{\partial t}\right)f=0 \tag{2.2.20}$$

Using a separation approach, let us assume that$l=\sqrt{K}\frac{\partial f}{\partial x}-\frac{\partial f}{\partial t}$. The original equation then becomes

$$K\frac{\partial^2 f}{\partial x^2}=\frac{\partial^2 f}{\partial t^2}\Longrightarrow\left(\sqrt{K}\frac{\partial}{\partial x}-\frac{\partial}{\partial t}\right)\left(\sqrt{K}\frac{\partial}{\partial x}+\frac{\partial}{\partial t}\right)f=0 \tag{2.2.21}$$

Then,

$$\sqrt{K}\frac{\partial l}{\partial x}+\frac{\partial l}{\partial t}=0 \tag{2.2.22}$$

From the solution to the transport equation, we obtain

$$l=L\left(x-\sqrt{K}t\right) \tag{2.2.23}$$

Similarly, by assuming that$m=\sqrt{K}\frac{\partial f}{\partial x}+\frac{\partial f}{\partial t}$, we obtain

$$m=M\left(-x-\sqrt{K}t\right) \tag{2.2.24}$$

At this point, it is important to clarify that $\sqrt{K}=v$, where v denotes the wave propagation speed. Absorbing the minus sign into the definition of the function M, and adding l and m, we obtain

$$\frac{2v\partial f}{\partial x}=L(x-vt)+M(x+vt) \tag{2.2.25}$$

Integrating both sides with respect to x, we obtain

$$f=\frac{1}{2v}\int[L(x-vt)+M(x+vt)]dx \tag{2.2.26}$$

Similarly, taking the difference between l and m, we obtain

$$\frac{2\partial f}{\partial t}=M(x+vt)-L(x-vt) \tag{2.2.27}$$

Integrating both sides with respect to t, we obtain

$$f=\frac{1}{2}\int[M(x+vt)-L(x-vt)]dt \tag{2.2.28}$$

For convenience, let us introduce the following definitions:

$$\begin{cases}\int L(\tau)d\tau=F(\tau)\\ \int M(\tau)d\tau=G(\tau)\end{cases} \tag{2.2.29}$$

Continuing the derivation, we obtain

$$\begin{cases} f = \dfrac{1}{2v}\int [L(x-vt) + M(x+vt)]dx = \dfrac{1}{2v}[F(x-vt) + G(x+vt)] + \theta(x) \\ f = \dfrac{1}{2}\int [M(x+vt) - L(x-vt)]dt = \dfrac{1}{2v}[F(x+vt) + G(x+vt)] + \rho(t) \end{cases} \tag{2.2.30}$$

The two expressions above must be equal. This requires that $\theta(x) = \rho(t)$. Since x and t are independent variables, both sides must therefore be equal to a constant $\theta(x) = \rho(t) = C$. Thus, the general solution to the wave equation is

$$f = \frac{1}{2v}[F(x-vt) + G(x+vt)] + C \tag{2.2.31}$$

For the wave equation (2.2.9), the propagation speed v is equal to the speed of light c. We may consider a particular solution by setting $G(x+vt)$ and C as zero. Furthermore, absorbing the factor $1/2v$ on the right-hand side of (2.2.30) into the definition of the function F, we obtain a simpler form of the wave equation solution:

$$f = F(x-ct) \tag{2.2.32}$$

Equation (2.2.31) tells us that an electromagnetic wave is a signal propagating outward at the speed of light c. Suppose that the function has a value f_0 at position x_0 and time t. After a time interval Δt, the function takes the same value f_0 at the position $x_0 + c \cdot \Delta t$. The functions F and G correspond to the two directions of wave propagation: F describes propagation in the positive x-direction, whereas G describes propagation in the negative x-direction.